

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ , then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

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Approaches tried

- **Angle-coordinate game:** Normalize 180° to 1. Cutting from angle A and splitting it as $x + (A - x)$ in a triangle (A, B, C) yields children $(B, x, 1 - B - x)$ and $(C, A - x, B + x)$.

This exact transition model underlies both directions.

- **Finite rational-grid attractor computation:** For targets p/q and grid-restricted states/cuts through $q = 20$, all states were winning exactly when $p \mid q$, suggesting the answer $t = 1/n$. This was exploratory only; arbitrary real cuts required a proof independent of computation.
- **Reciprocal-target construction:** For $t = 1/n$, complementary angles at the cut point can be chosen as m/n and $(n - m)/n$. An interval argument finds an admissible m for every starting triangle; then an induction reduces any retained angle k/n to $1/n$.
- **Safe-family invariant for necessity:** For nonreciprocal t , call a triangle safe when no angle is a positive integral multiple of t . An exhaustive four-case use of the transition formula proves that every cut of a safe triangle has a safe child. An equilateral starting triangle is safe, so Shan-Yu can avoid the target forever.
- **Earlier acute-triangle idea:** For $t > 1/2$, retaining an acute child gives a special-case avoidance strategy, but the safe-family invariant supersedes it and handles all nonreciprocal targets uniformly.

Current best

The complete characterization is

$$\theta = \frac{180^\circ}{n} \quad \text{for some integer } n \geq 2.$$

For these and only these values, Mulan has a finite winning strategy. In fact, after at most one initialization cut she forces a positive integral multiple $k\theta$ in either possible child, and then needs at most $k - 1 \leq n - 2$ further cuts.

Full proof

Normalize all angle measures by dividing by 180° . Thus a triangle's angles sum to 1, and put

$$t = \frac{\theta}{180^\circ} \in (0, 1).$$

We prove that Mulan can force a win exactly when $t = 1/n$ for an integer $n \geq 2$.

1. The transition rule

Suppose the current triangle has angles (A, B, C) . Mulan cuts from the vertex of angle A to the opposite side, and let x be the portion of angle A lying in the child containing the vertex of angle B . Since the cut is interior,

$$0 < x < A.$$

The two children have angle triples

$$(B, x, 1 - B - x) \quad (1)$$

and

$$(C, A - x, B + x). \quad (2)$$

Indeed, the first two angles of the first child are B, x , so its third angle is $1 - B - x$. The second child has angles $C, A - x$ at old vertices, so its angle at the cut point is

$$1 - C - (A - x) = B + x.$$

Conversely, every $x \in (0, A)$ is attainable: the unique ray inside the angle A making angle x with the appropriate side meets the opposite side at an interior point.

2. Sufficiency when $t = 1/n$

Let $n \geq 2$ be an integer and suppose $t = 1/n$. We first show that, from any triangle not already containing angle t , Mulan can make a cut after which either possible child contains an angle k/n for some integer $k \in \{1, \dots, n-1\}$.

Let the triangle's angles be (A, B, C) .

If $n = 2$, then at least two angles are less than $1/2$: otherwise either one is exactly $1/2$, in which case Mulan has already won, or two are greater than $1/2$, which is impossible because the angle sum is 1. Label two angles below $1/2$ as B, C and the third as A . Then

$$C < \frac{1}{2} < 1 - B = A + C. \quad (3)$$

Now let $n \geq 3$, and choose A to be a largest angle. Then $A \geq 1/3$. In fact,

$$nA > 1. \quad (4)$$

For $n > 3$ this follows from $nA \geq n/3 > 1$. For $n = 3$, if $3A = 1$, then maximality and the angle sum force $A = B = C = 1/3 = t$, contrary to the assumption that the triangle does not already contain the target.

The open interval

$$(nC, n(A + C))$$

has length $nA > 1$, so it contains an integer m . To justify this explicitly, choose the least integer strictly greater than nC ; it is at most $nC + 1 < n(A + C)$. Moreover,

$$0 < nC < m < n(A + C) = n(1 - B) < n,$$

so $m \in \{1, \dots, n-1\}$. Consequently,

$$C < \frac{m}{n} < A + C. \quad (5)$$

For $n = 2$, the same conclusion holds with $m = 1$ by (3).

Mulan now cuts from the vertex with angle A , choosing

$$x = A + C - \frac{m}{n} = 1 - B - \frac{m}{n}.$$

The right inequality in (5) gives $x > 0$, and the left inequality gives $x < A$, so this is a legal cut. By (1) and (2), the angles at the new point in the two children are

$$1 - B - x = \frac{m}{n} \quad \text{and} \quad B + x = 1 - \frac{m}{n} = \frac{n - m}{n}. \quad (6)$$

Both are positive integral multiples of $t = 1/n$. Hence whichever child Shan-Yu retains has an angle kt with $1 \leq k \leq n-1$.

It remains to show that any such multiple can be reduced to the target. We use induction on k . If $k = 1$, Mulan has already won. If $k \geq 2$, she cuts from the vertex whose angle is kt , using the interior ray which splits it into

$$t + (k-1)t.$$

This is legal because both portions are positive. One child contains angle t , while the other contains angle $(k-1)t$. Thus Shan-Yu either retains a triangle with the target angle, or leaves a strictly smaller positive integral multiple, to which the induction hypothesis applies. Mulan therefore wins after finitely many cuts. This proves sufficiency.

3. Necessity for all other t

Suppose now that

$$t \neq \frac{1}{n} \quad \text{for every integer } n \geq 2. \quad (7)$$

Call a triangle **safe** if none of its angles is a positive integral multiple of t .

Shan-Yu can choose an equilateral initial triangle, and it is safe. Indeed, if its angle $1/3$ were kt for a positive integer k , then $t = 1/(3k)$, contrary to (7).

We claim that after any legal cut of a safe triangle, at least one child is safe. Let the safe parent have angles (A, B, C) , and suppose the cut is from the vertex of angle A . Use the parameter $x \in (0, A)$ from (1)–(2). Assume for contradiction that both children are unsafe.

The inherited angle B in the first child is not a positive integral multiple of t , because the parent is safe. Therefore one of the other two angles of that child must be such a multiple: for some positive integer r ,

$$x = rt \quad \text{or} \quad 1 - B - x = rt. \quad (8)$$

Likewise, inherited angle C in the second child is not a multiple, so for some positive integer s ,

$$A - x = st \quad \text{or} \quad B + x = st. \quad (9)$$

There are four possibilities.

- If $x = rt$ and $A - x = st$, then $A = (r+s)t$, contradicting safety of the parent.
- If $x = rt$ and $B + x = st$, then $B = (s-r)t$. As $B > 0$, we have $s-r \geq 1$, again contradicting safety.
- If $1 - B - x = rt$ and $A - x = st$, then

$$C = 1 - A - B = (1 - B - x) - (A - x) = (r-s)t.$$

Since $C > 0$, we have $r-s \geq 1$, contradicting safety.

- If $1 - B - x = rt$ and $B + x = st$, adding gives $1 = (r+s)t$. Hence $t = 1/(r+s)$. Since $r+s \geq 2$, this contradicts (7).

All four cases are impossible, proving the claim. Shan-Yu can therefore retain a safe child after every cut. Every retained triangle remains safe, and in particular never has an angle equal to t . Thus Mulan cannot guarantee a win when (7) holds.

Combining the two directions and restoring degrees gives exactly

$$\theta = \frac{180^\circ}{n} \quad (n = 2, 3, 4, \dots).$$